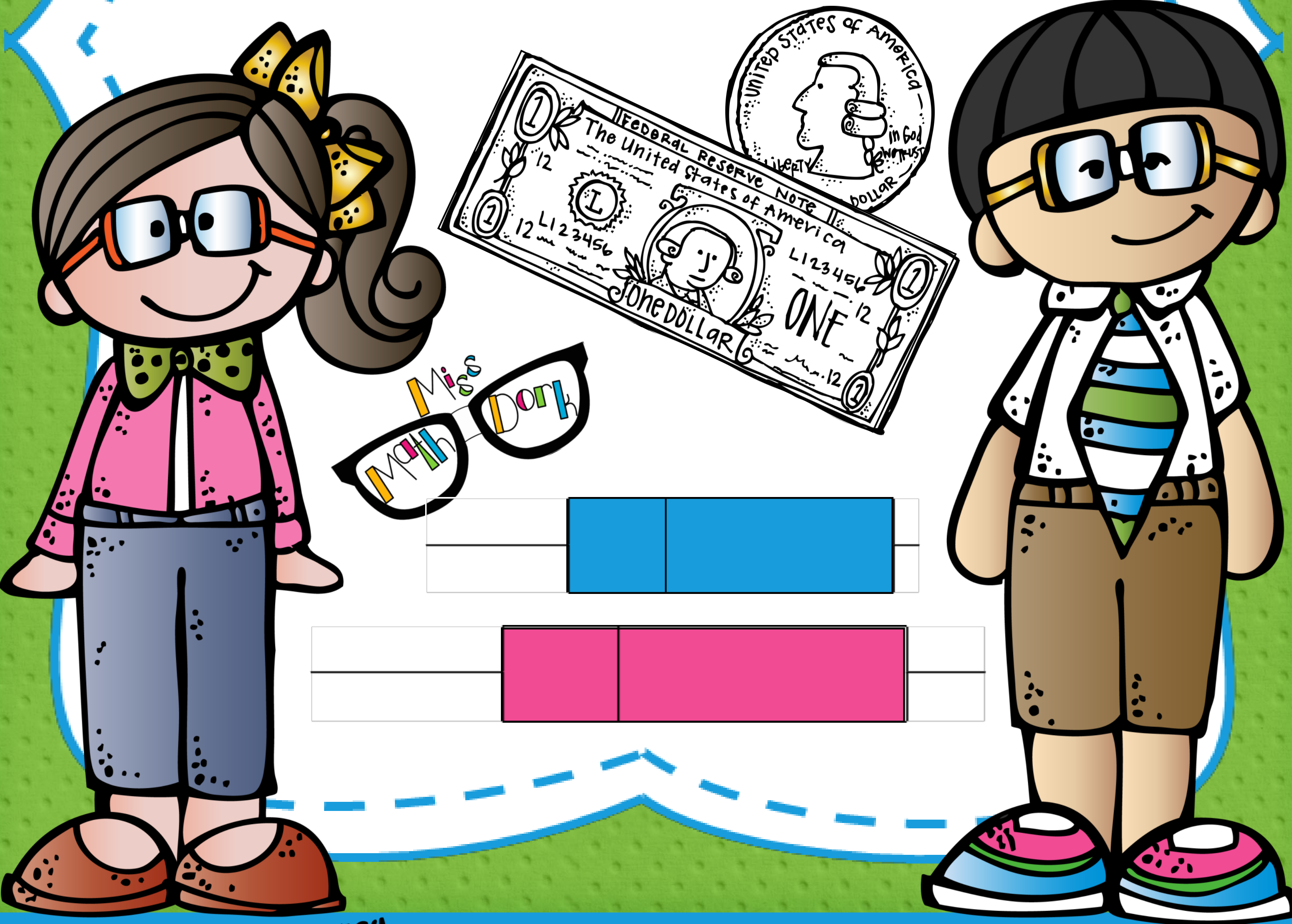
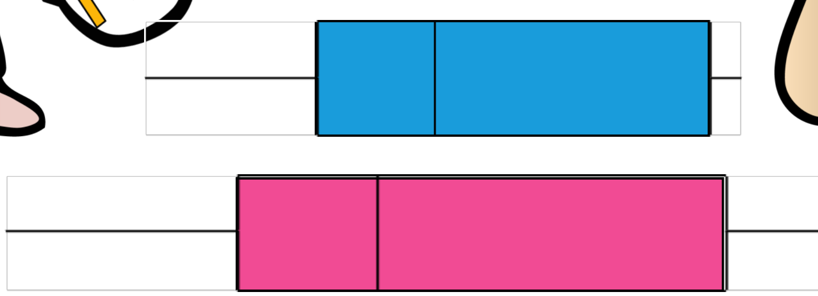
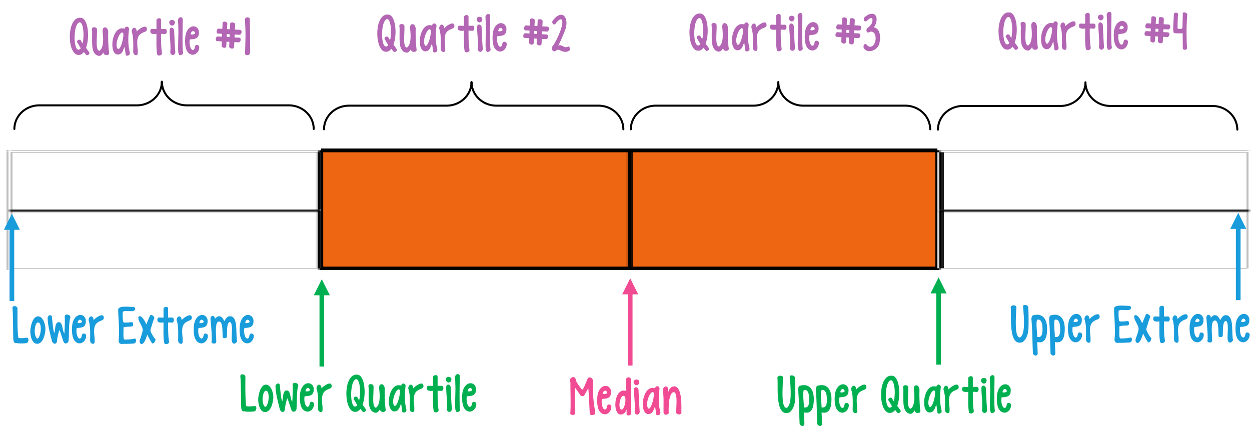
# UNDERSTANDING, IDENTIFYING & ANALYZING BOX & WHISKER PLOTS!



# UNDERSTANDING THE PARTS OF BOX & WHISKER PLOTS

A box and whisker plot is a way to organize and display large amounts of data by breaking the data into four (4) quartiles at five (5) key data points.

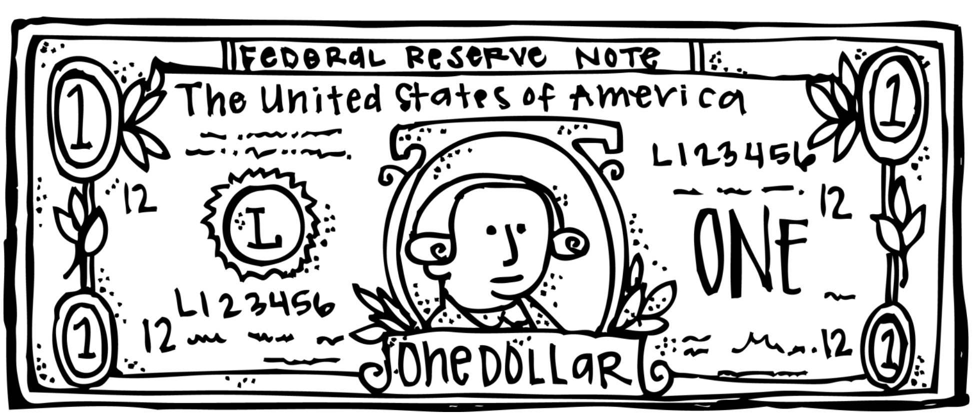



Each quartile of the box and whisker plot contains the 1/4 or 25 % of the collective data.

The "box" of the box and whisker plot is the middle 50% of data and has a special name, the inter quartile range (IQR). The IQR can be calculated by subtracting the Lower Quartile from the Upper Quartile. IQR = Upper Quartile - Lower Quartile
Located in the IQR is the middle piece of data called the Median.

The "whiskers" of the box and whisker plot are the upper and lower 25% of the data.
The smallest point of data is called the Lower Extreme.
The largest point of data is called the Upper Extreme.

The box and whisker above has 4 quartiles that are all similar in size. This means the data is evenly distributed. Because we can't actually see the data, sometimes it's hard to picture exactly how everything is spread out. That's when it's great to compare a box and whisker plot and a one dollar bill.



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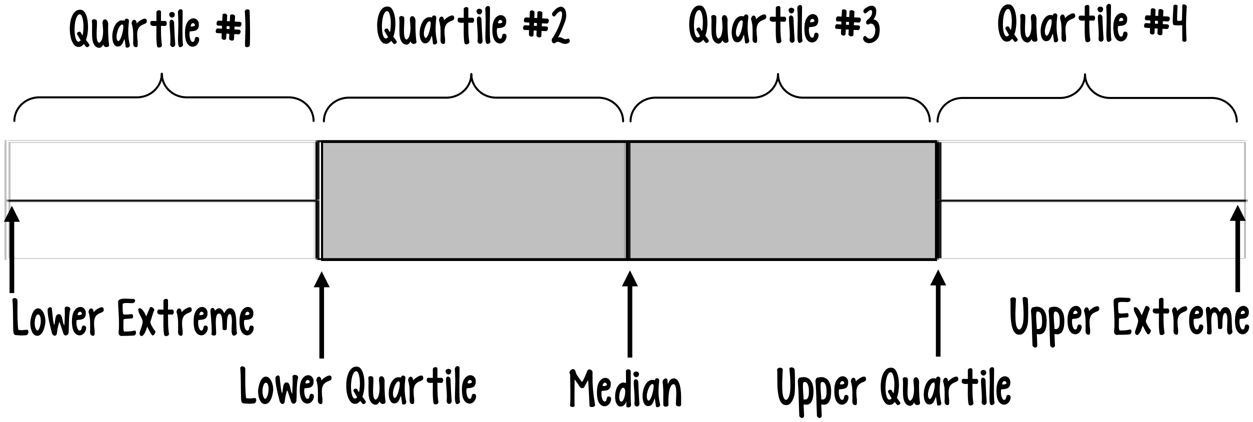


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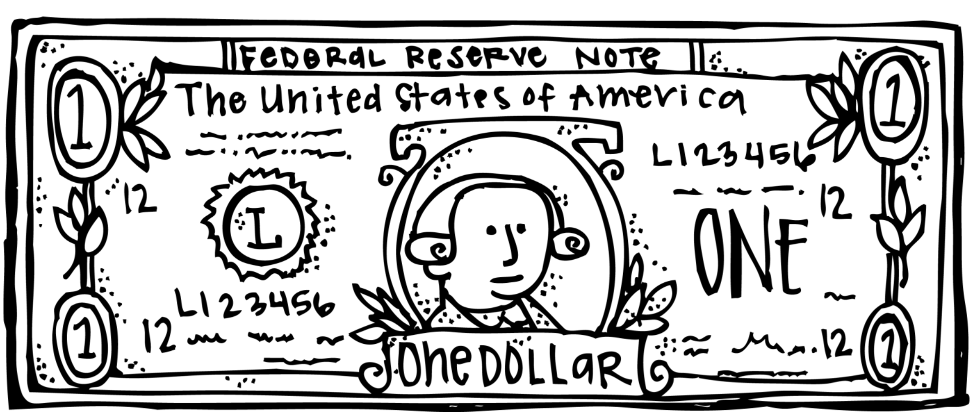
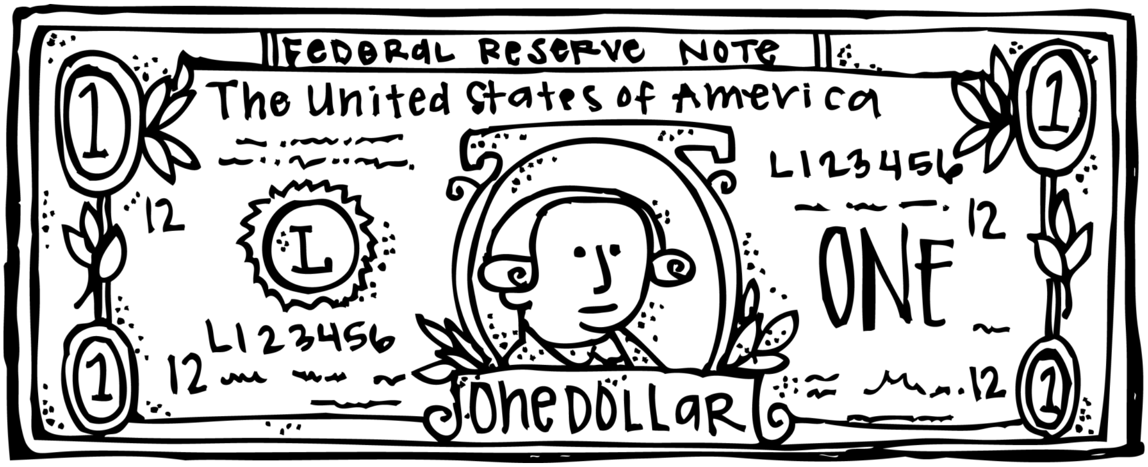
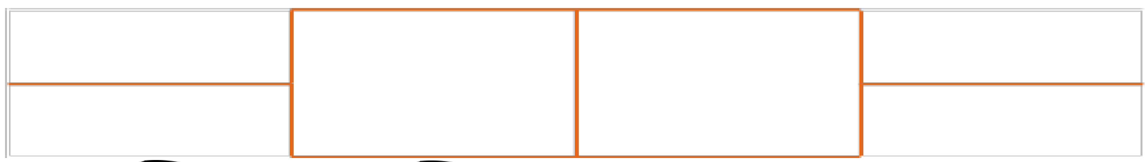
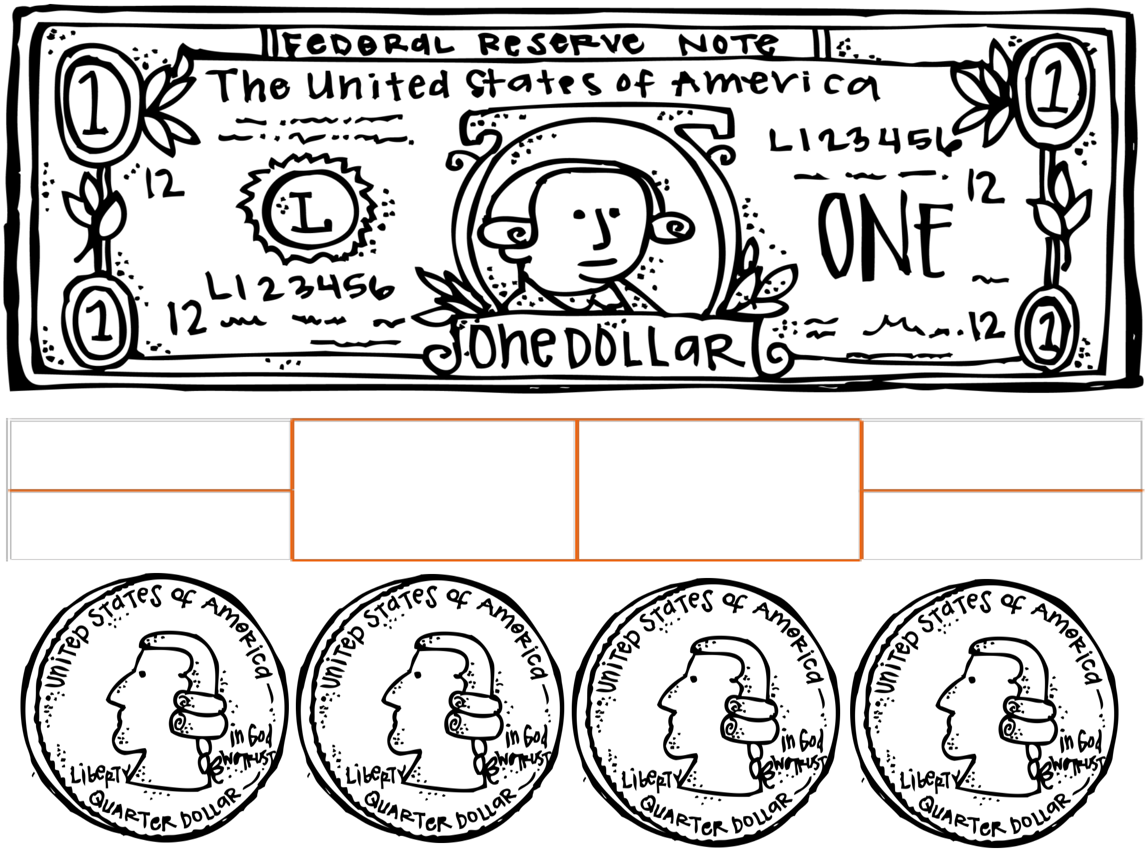


# BOX & WHISKER PLOTS AND THE DOLLAR BILL!




# BOX & WHISKER PLOTS AND THE DOLLAR BILL! PG 1




How much is each coin worth?

What percent of data is in each quartile?

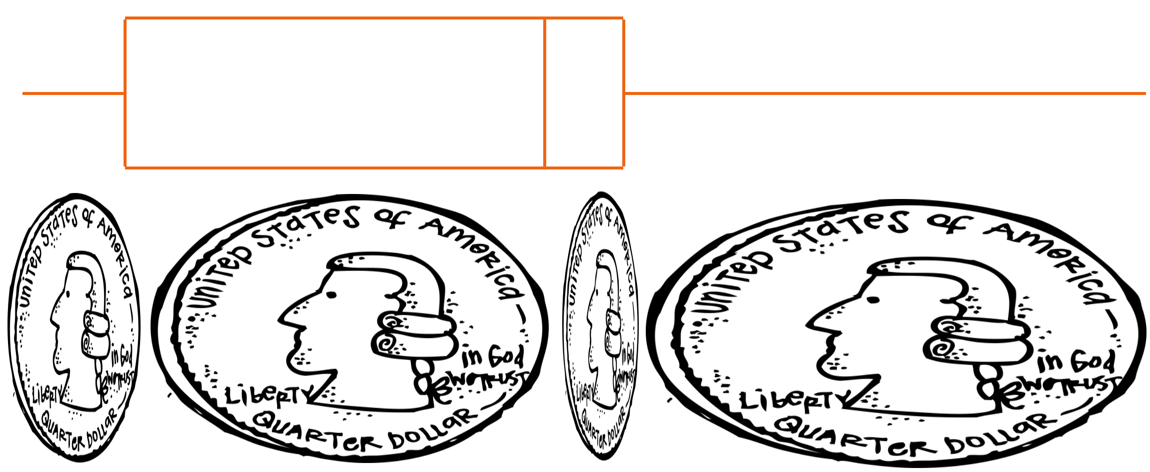

Now that the coins have been distorted, how much is each worth? Did the value change for the coin?

Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile?

# BOX & WHISKER PLOTS AND THE DOLLAR BILL! PG 2

Now that the coins have been distorted, how much is each worth? Did the value change for the coin?

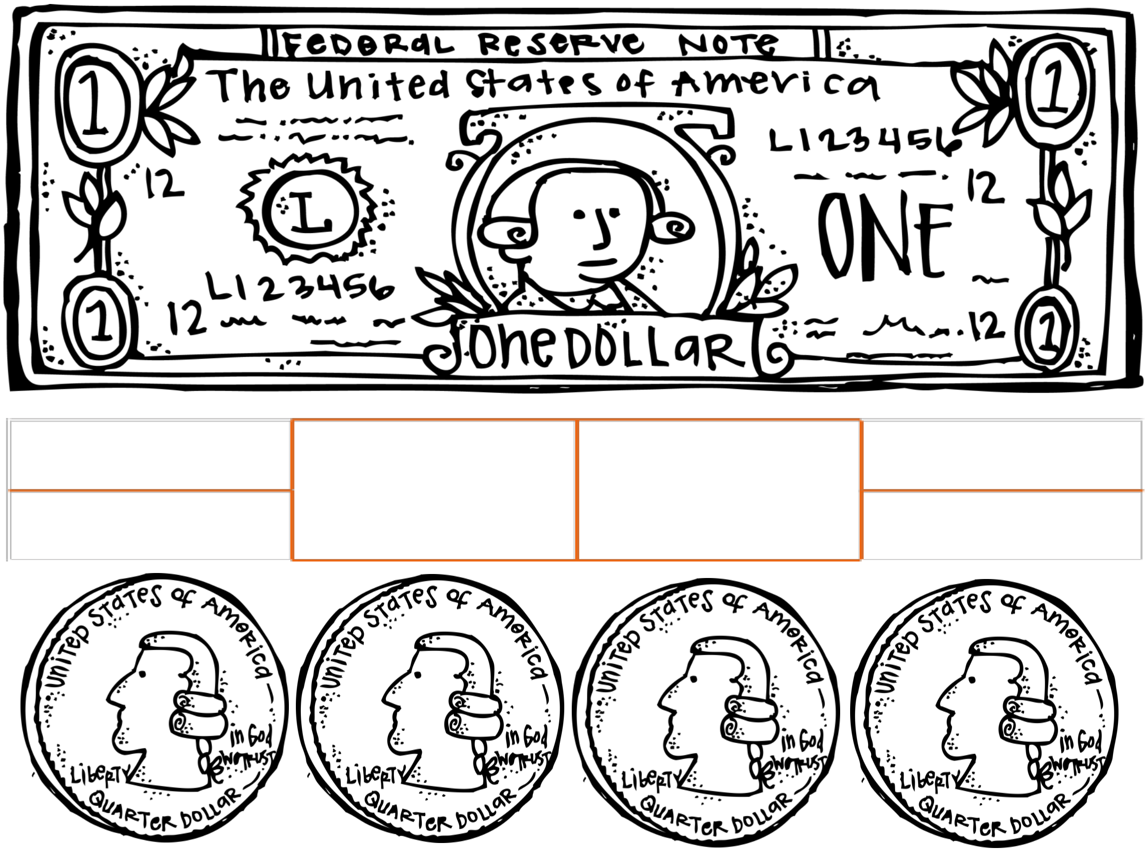
Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile?



Now that the coins have been distorted, how much is each worth? Did the value change for the coin?

Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile?

# BOX & WHISKER PLOTS AND THE DOLLAR BILL! KEY I




How much is each coin worth? 25 cents

What percent of data is in each quartile? 25 percent

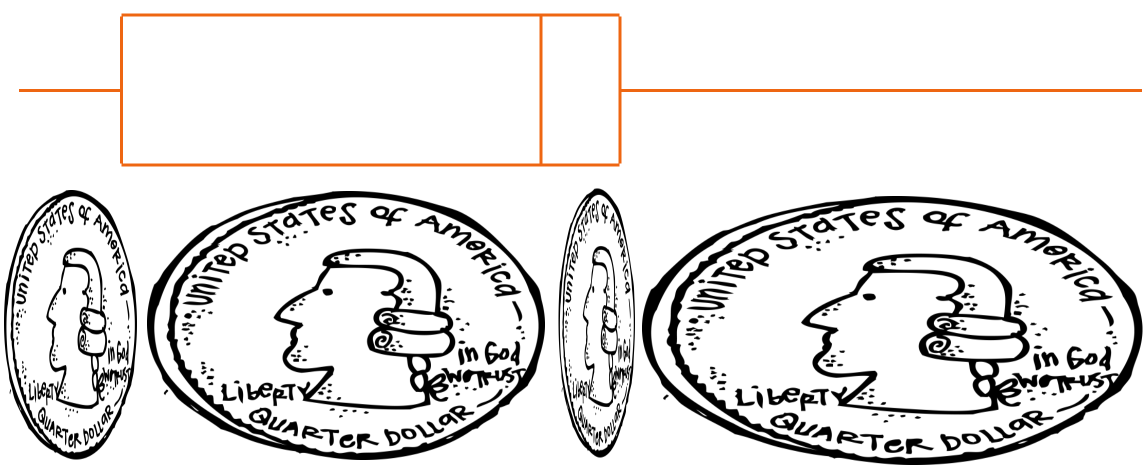
Now that the coins have been distorted, how much is each worth? Did the value change for the coin? Still worth 25 cents each. The value did not change

Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile? Still worth 25 percent each. The value did not change

# BOX & WHISKER PLOTS AND THE DOLLAR BILL! KEY 2

Now that the coins have been distorted, how much is each worth? Did the value change for the coin? Still worth 25 cents each. The value did not change

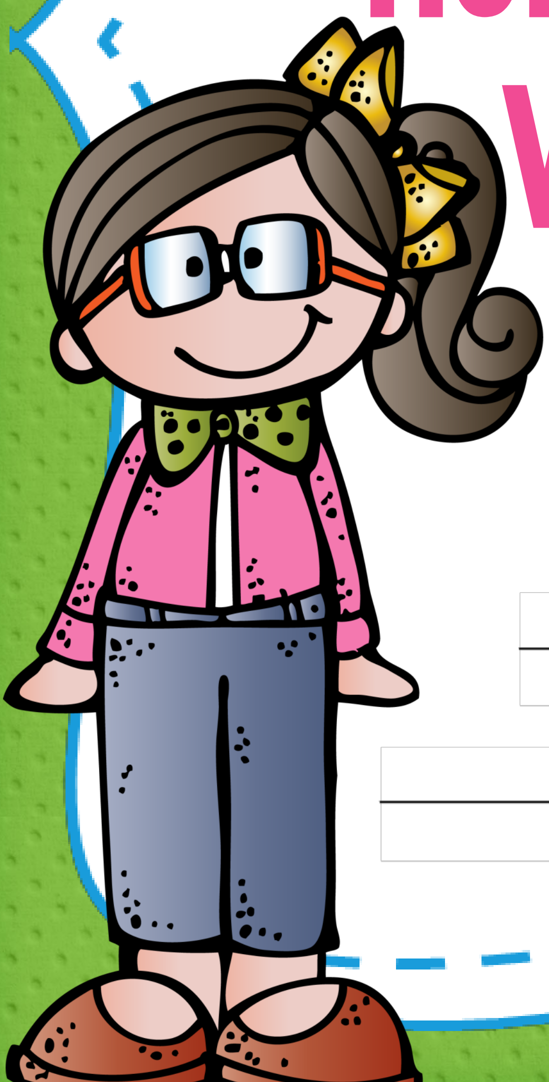
Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile? Still worth 25 percent each. The value did not change

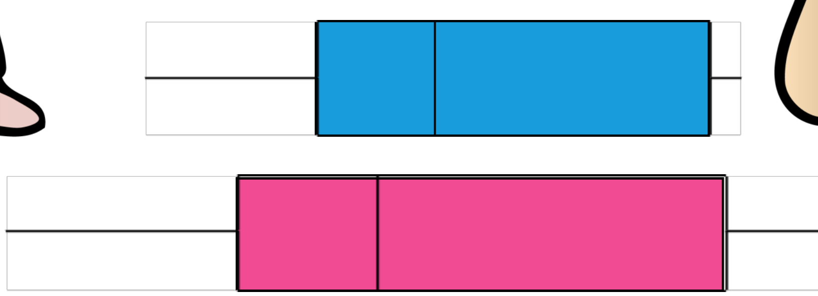


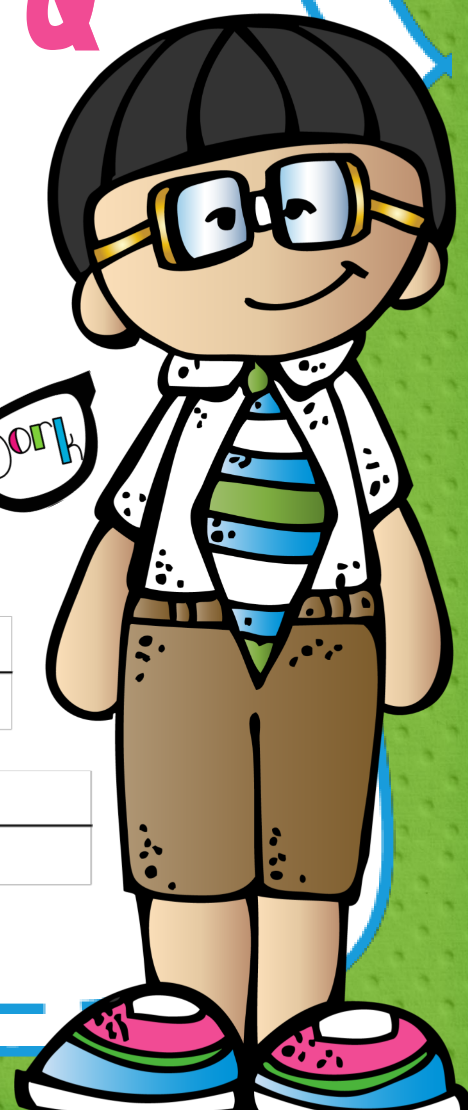
Now that the coins have been distorted, how much is each worth? Did the value change for the coin? Still worth 25 cents each. The value did not change

Now that the quartiles have been distorted, what percent of data is in each quartile? Did the value change for the quartile? Still worth 25 percent each. The value did not change

# UNDERSTANDING DATA SPREAD & CREATING A HUMAN BOX & WHISKER PLOT







# BOX & WHISKER PLOTS KNOWN & UNKNOWN VALUES

When working with box and whisker plots, the number of data pieces in your set can help determine certain pieces of data. There are 4 different ways the data can fall:
X remainder 0, X remainder 1, X remainder 2, X remainder 3.

The "X" is the number of data points in each quartile. The "remainder" helps determine which key data points we know the exact value of.

Remainder 0 = Median, Lower and Upper Quartiles are averaged.
Remainder 1 = Exact Median is known. Lower and Upper Quartiles are averaged.
Remainder 2 = Median is averaged. Lower and Upper Quartiles are known.
Remainder 3 = Median, Lower and Upper Quartiles are known.

| # of data points | Data breakdown | Example with different data points |
|---|---|---|
| 12 | 3 R0 | 3 pieces of data in each quartile. Lower Quartile, Median and Upper Quartile all averages. |
| 13 | 3 R1 | 3 pieces of data in each quartile. Lower and Upper quartile averages, Median is the 13th data point. |
| 14 | 3 R2 | 3 pieces of data in each quartile. Median is an average, Lower and Upper Quartiles are the 13th and 14th data point. |
| 15 | 3 R3 | 3 pieces of data in each quartile. Lower Quartile, Median and Upper Quartile are the 13th, 14th, and 15th piece of data. |

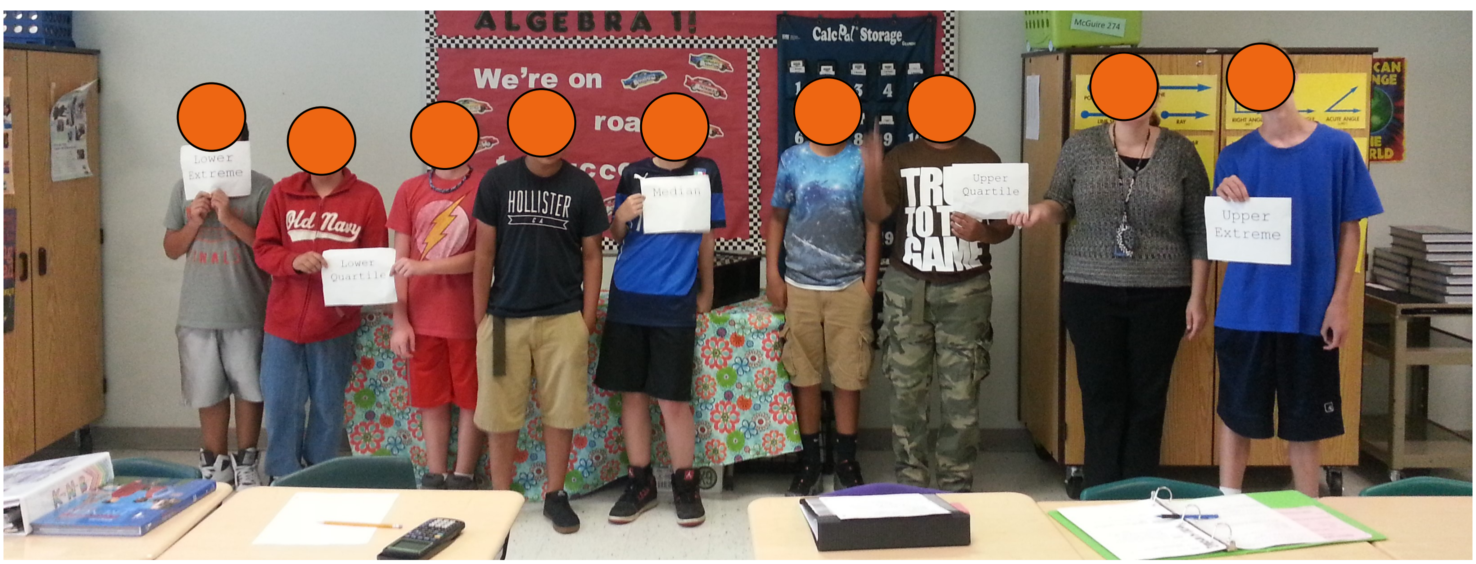
# HUMAN BOX & WHISKER PLOTS

After students have started visualizing squished and stretched quarters it's a good idea to get them up and moving! Let's make a human box and whisker plot!

Print the next 5 pages on sturdy paper, or laminate / put in a page protector. Use the "**BOX & WHISKER PLOTS KNOWN & UNKNOWN VALUES**" page to help students decide where to hold the paper.

Have students line up given a scenario (see a few examples below). Once they have lined up, have them find each of the 5 key data points. The student(s) who identifies with the data point should hold the corresponding paper. If the data point is an average, two students will hold the paper between themselves.

Remember the data needs to be Quantitative (non-categorical):
Height (short to tall), # of jumping jacks in 30 seconds, grade on a silly quiz, etc.




# Lower Extreme

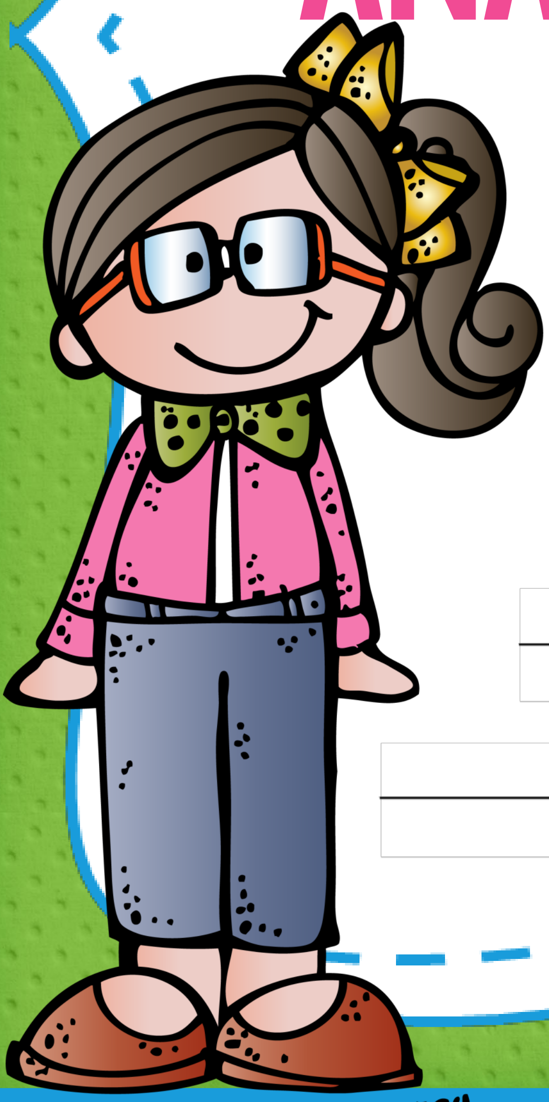
# Lower Quartile

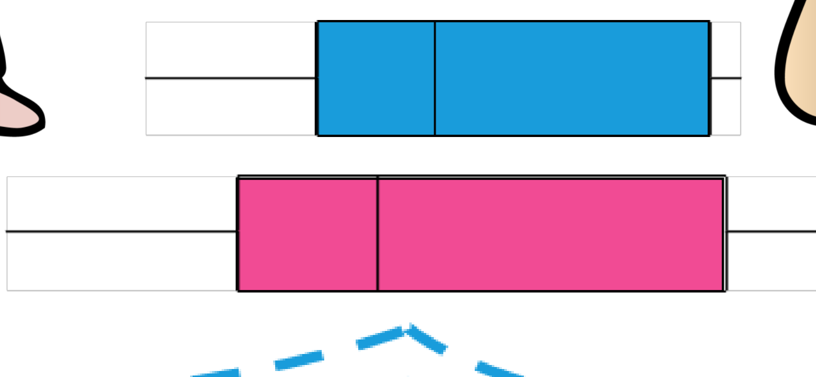
# Median

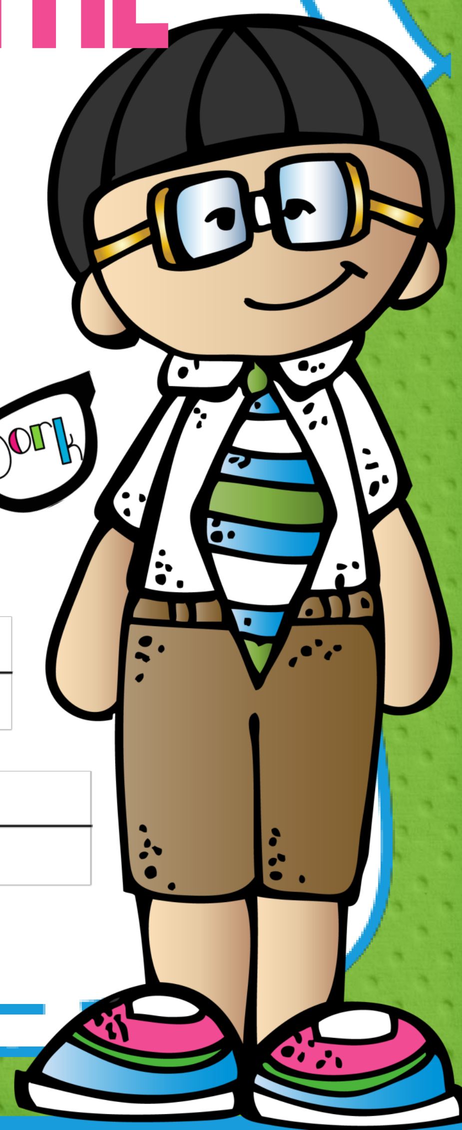
# Upper Quartile

# Upper Extreme

# IDENTIFYING BOX & WHISKER VOCABULARY THEN ANALYZING THE DATA





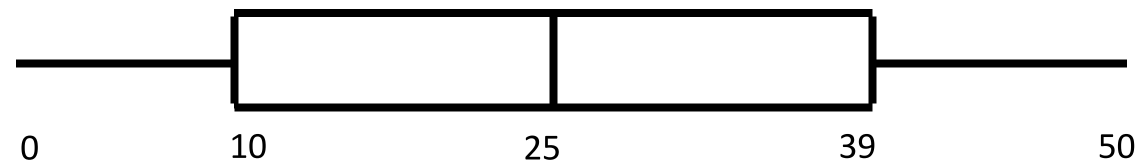


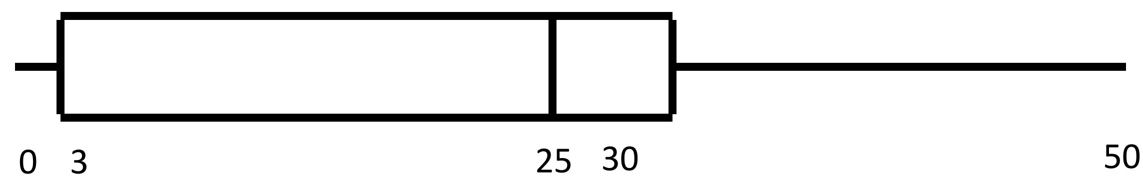
# IDENTIFYING BOX & WHISKER PLOTS W/ COLOR

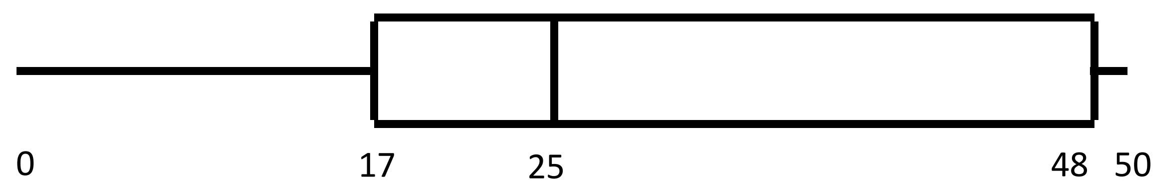
In order to analyze box and whisker plots, you must first be able to easily identify all the parts. Use the given key to add some color to the box and whisker plots below.

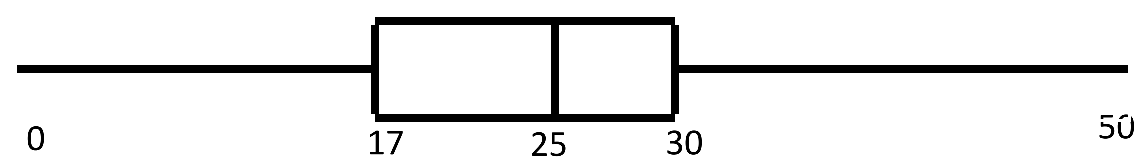
COLOR CODE KEY

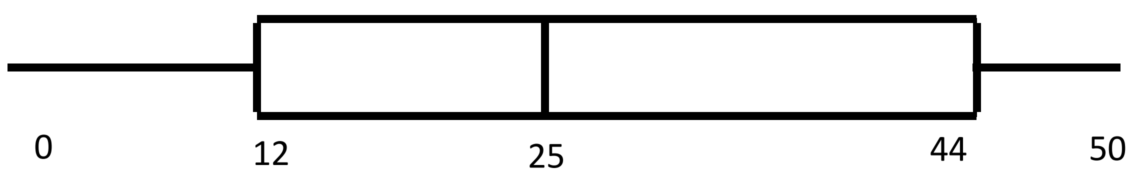
| | |
|---|---|
| 5 key values = BLUE & LABEL | 3rd Quartile = PURPLE POLKA DOTS |
| 1st Quartile = PURPLE | 4th Quartile = ORANGE |
| 2nd Quartile = GREEN STRIPES | Inter-Quartile Range = YELLOW |

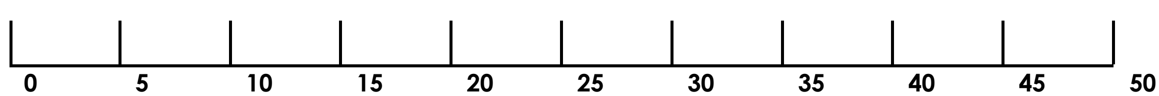


# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 1

Use the given key to add some color to the box and whisker plots below. After you have colored each box and whisker plot, answer the questions on the following page. Assume that all 5 sets represent the same amount of data.

COLOR CODE KEY

| | |
|---|---|
| 5 key values = BLUE & LABEL | 3rd Quartile = PURPLE POLKA DOTS |
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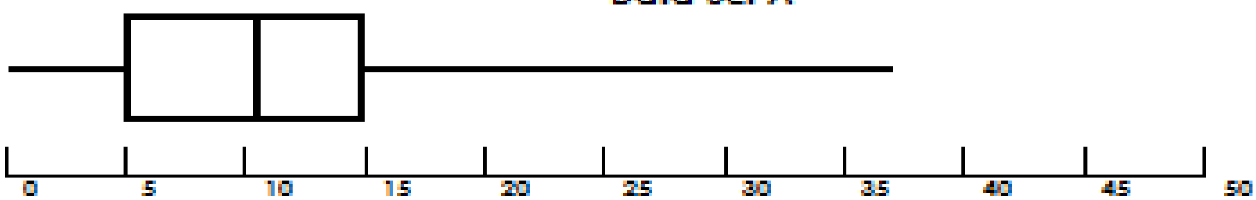
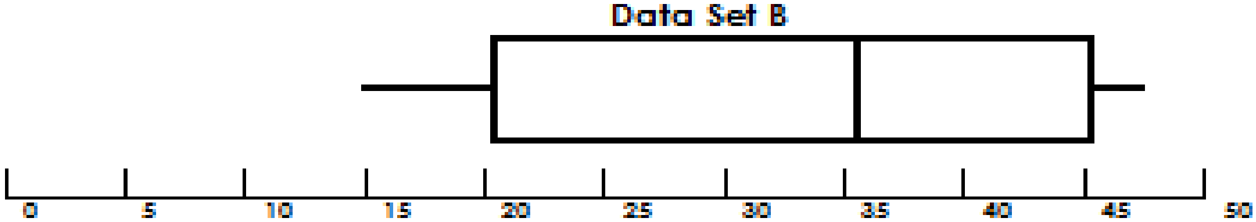
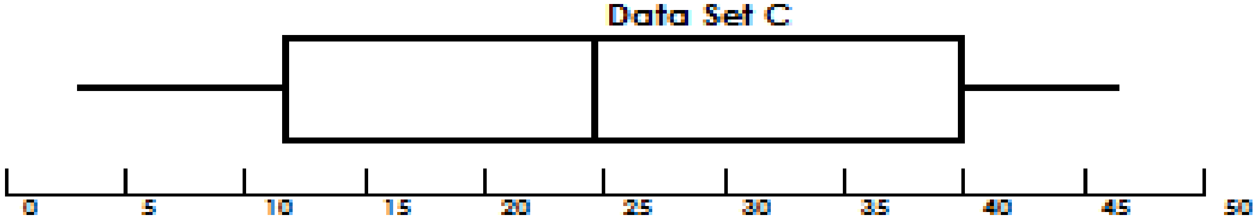
Data Set A



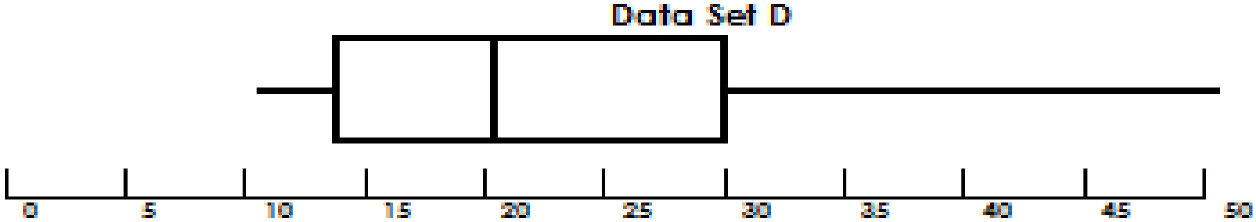

Data Set B



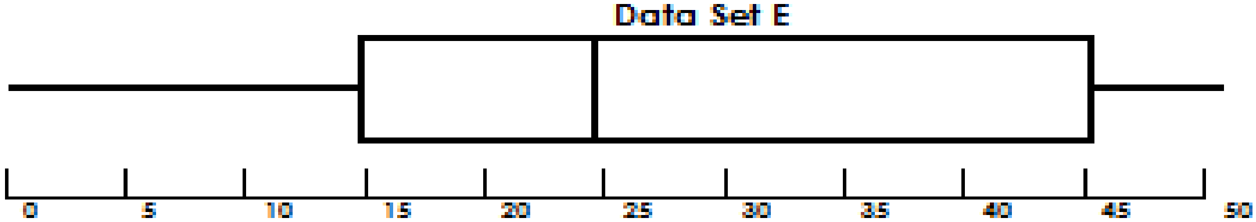

Data Set C




Data Set D




Data Set E



# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 2

1. In data set A, what percent of data is greater than 10 hours?

2. In data set B, what percent of data is less than 45 hours?

3. In data set C, what percent of data is at least 25 hours?

4. In data set D, what percent of data is at most 30 hours?

5. In data set E, what percent of data is at least 45 hours?

6. Comparing data set B and D, which has the higher percentage of students playing at least 20 hours?

7. Comparing data set C and E, which data set has more people playing at least 25 hours?

8. Comparing data sets A and E, which data set has more people playing at least 15 hours?

9. In which set of data, is the Inter Quartile Range (IQR) the largest?

10. In which set of data, is the Inter Quartile Range (IQR) the smallest?

11. In which set of data, is the Median the largest?

12. In which set of data, is the Median the smallest?

# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 3

Use the given key to add some color to the box and whisker plots below. After you have colored each box and whisker plot, answer the questions on the following page. Each data set has a DIFFERENT set of data per quartile.

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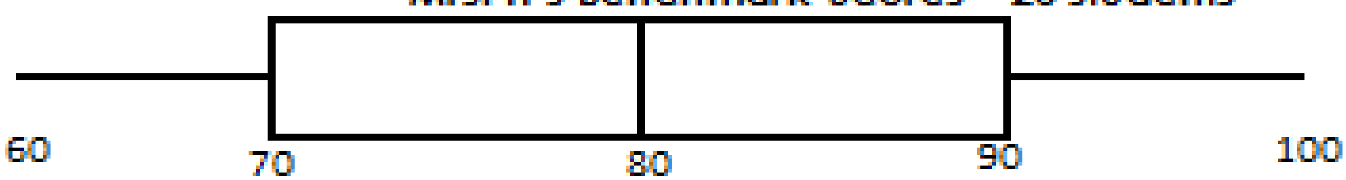
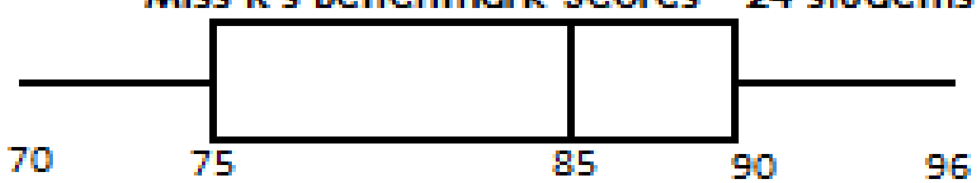
## Math Class Test Scores

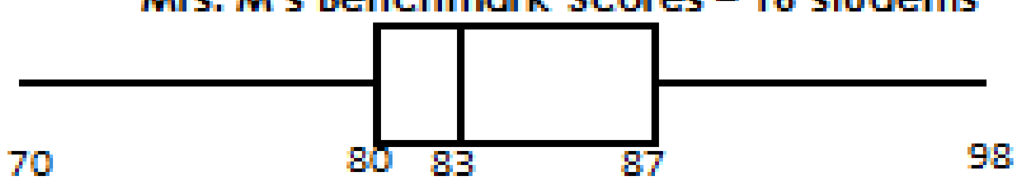
Mrs. H's Benchmark Scores – 20 students



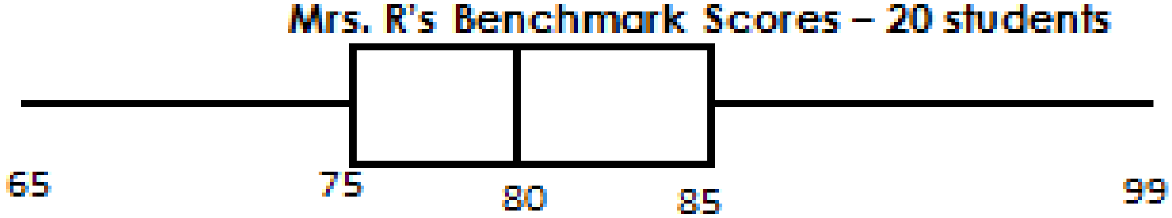

Miss K's Benchmark Scores – 24 students




Mrs. M's Benchmark Scores – 16 students




Mrs. R's Benchmark Scores – 20 students



# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 4

1. How many students make up 25% of Mrs. H's class?

2. How many students make up 25% of Miss K's class?

3. How many students make up 25% of Mrs. Ms class?

4. How many students make up 25% of Mrs. R's class?

5. Comparing Mrs. H's and Miss. K's classes: Which class has a greater number of student's scoring at least a 90 on their math test?

6. Comparing Mrs. M's and Mrs. R's classes: Which class has a greater number of student's scoring at most an 80 on their math test?

7. Which class has the highest median score?

8. Who has the most students scoring at least a 75 on their test? Miss K or Mrs. R?

# IDENTIFYING BOX & WHISKER PLOTS W/ COLOR KEY

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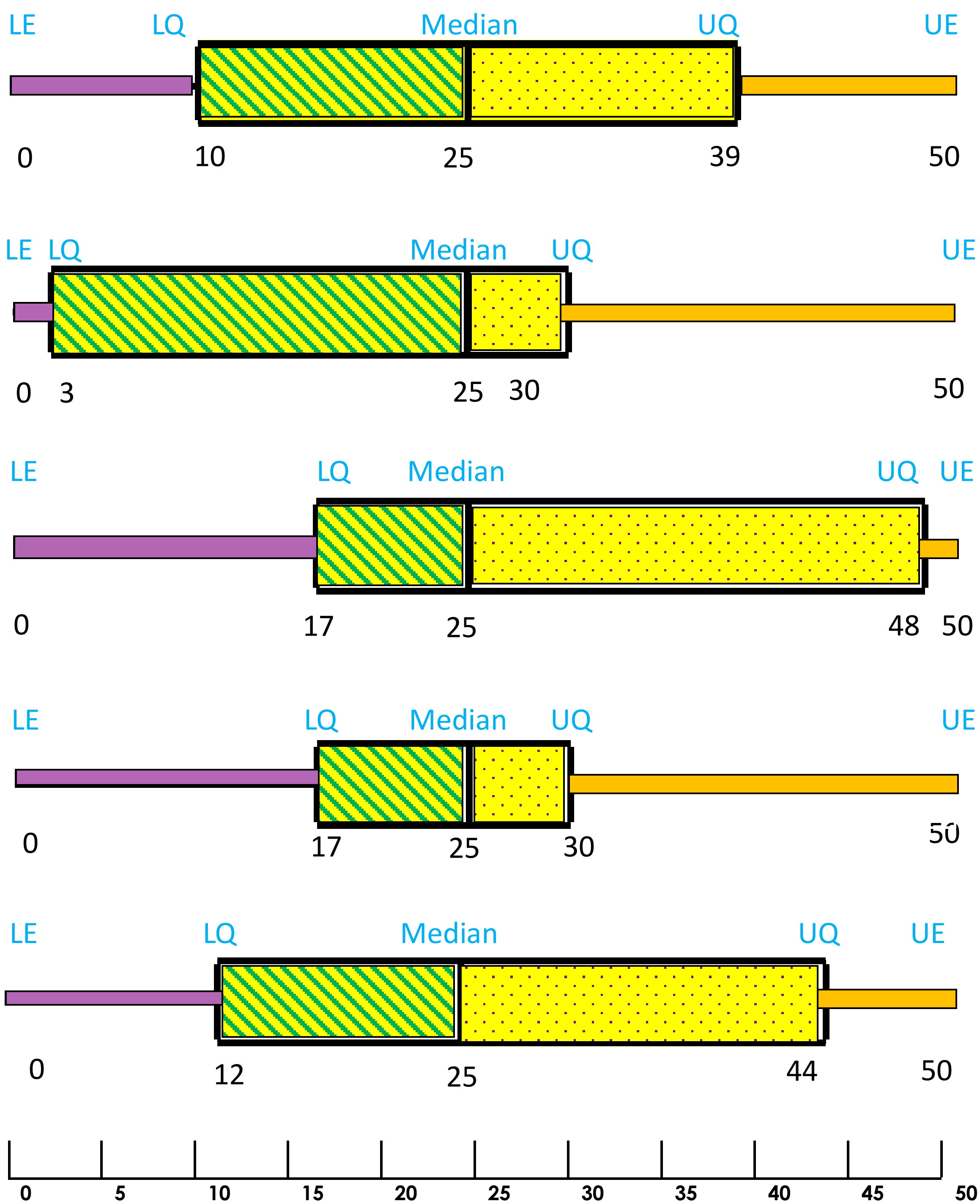

# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 1 KEY

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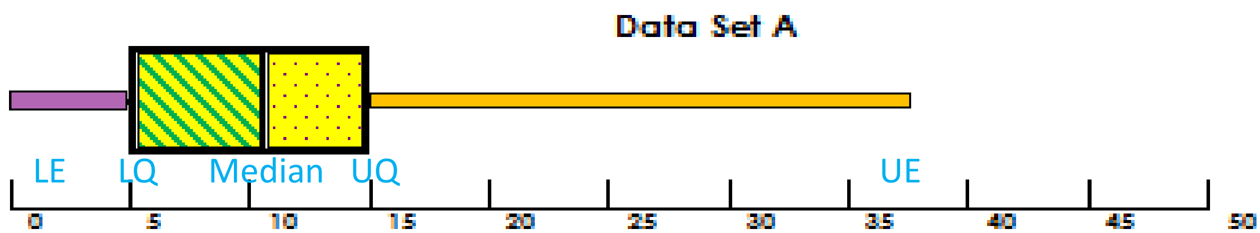
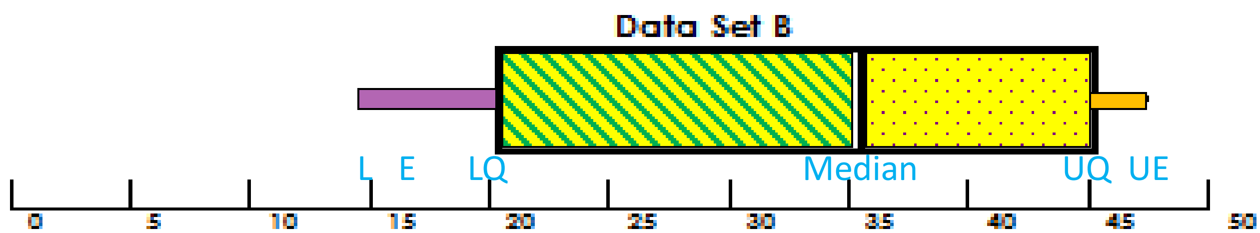
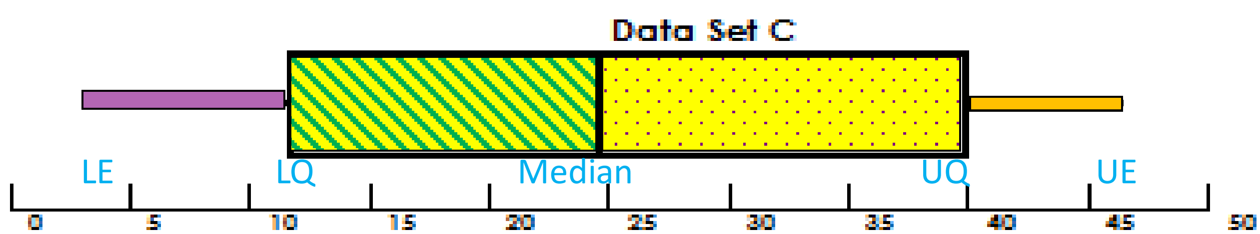
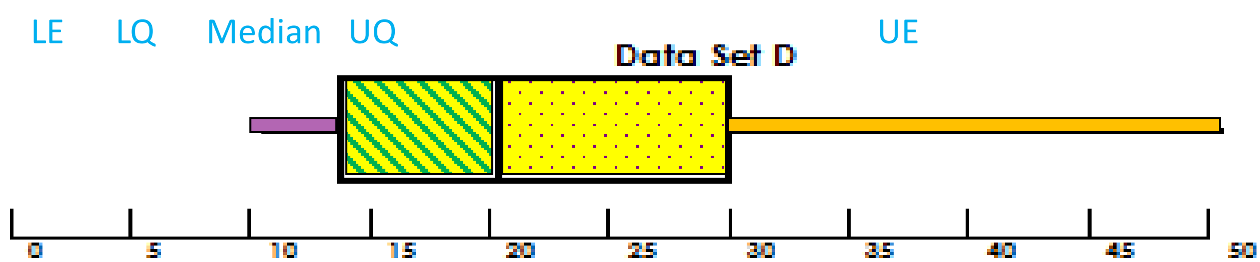
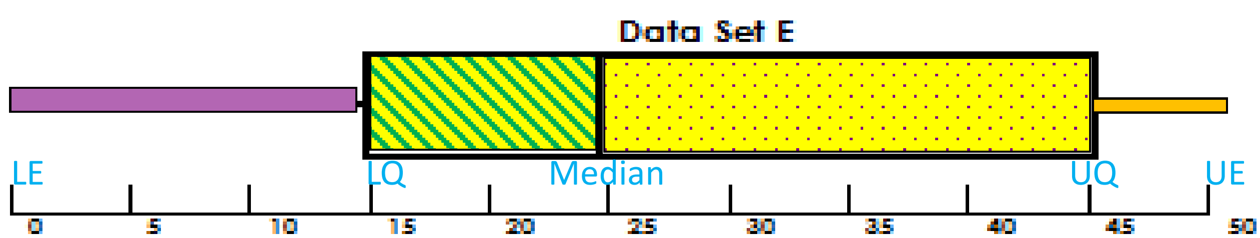


# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 2 KEY

1. In data set A, what percent of data is greater than 10 hours? 50%

2. In data set B, what percent of data is less than 45 hours? 75%

3. In data set C, what percent of data is at least 25 hours? 50%

4. In data set D, what percent of data is at most 30 hours? 75%

5. In data set E, what percent of data is at least 45 hours? 25%

6. Comparing data set B and D, which has the higher percentage of students playing at least 20 hours? Set B

7. Comparing data set C and E, which data set has more people playing at least 25 hours? They have the same amount

8. Comparing data sets A and E, which data set has more people playing at least 15 hours? Set E

9. In which set of data, is the Inter Quartile Range (IQR) the largest? Set E the IQR is 30

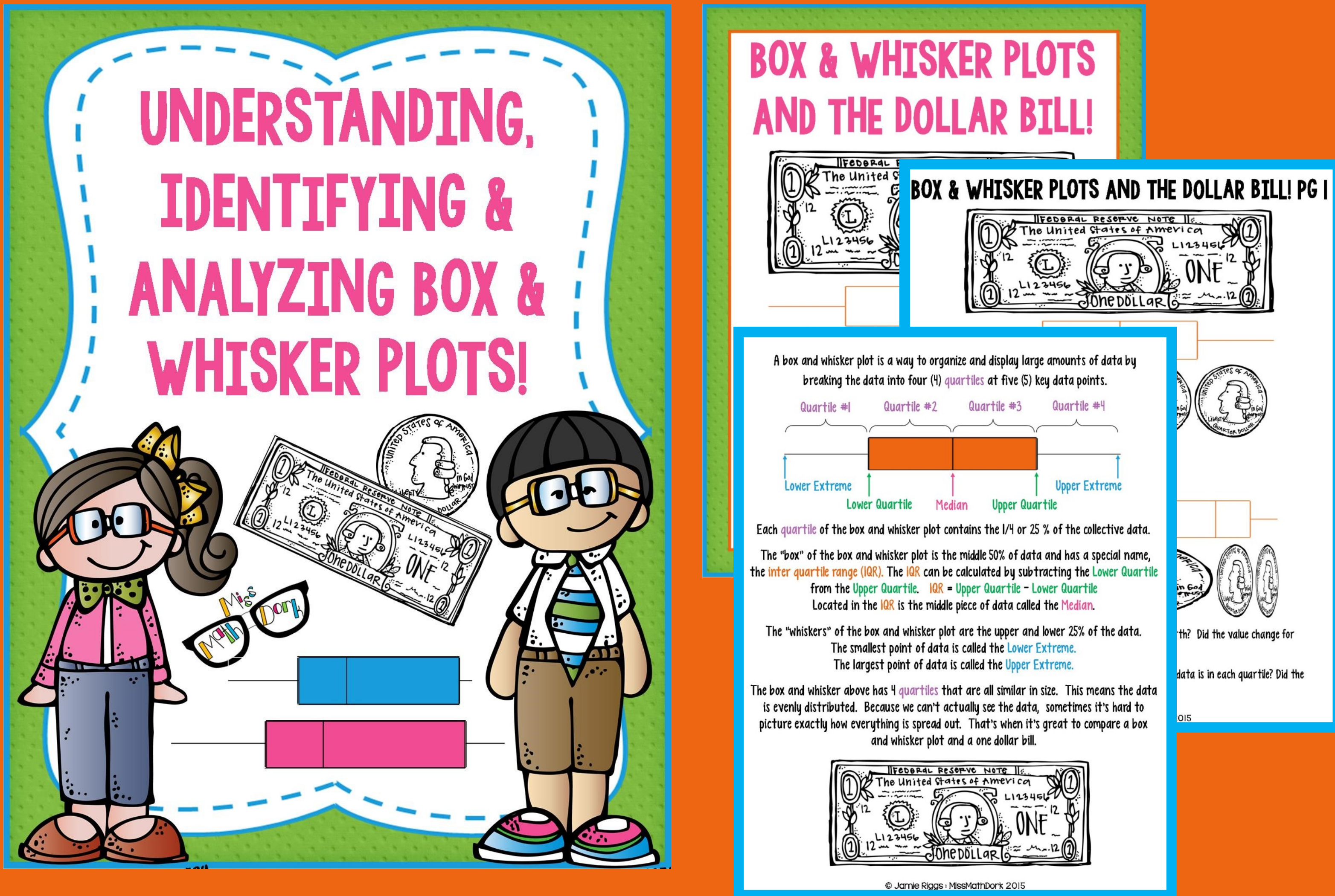
10. In which set of data, is the Inter Quartile Range (IQR) the smallest? Set A the IQR is 10

11. In which set of data, is the Median the largest? Set B the median is 36

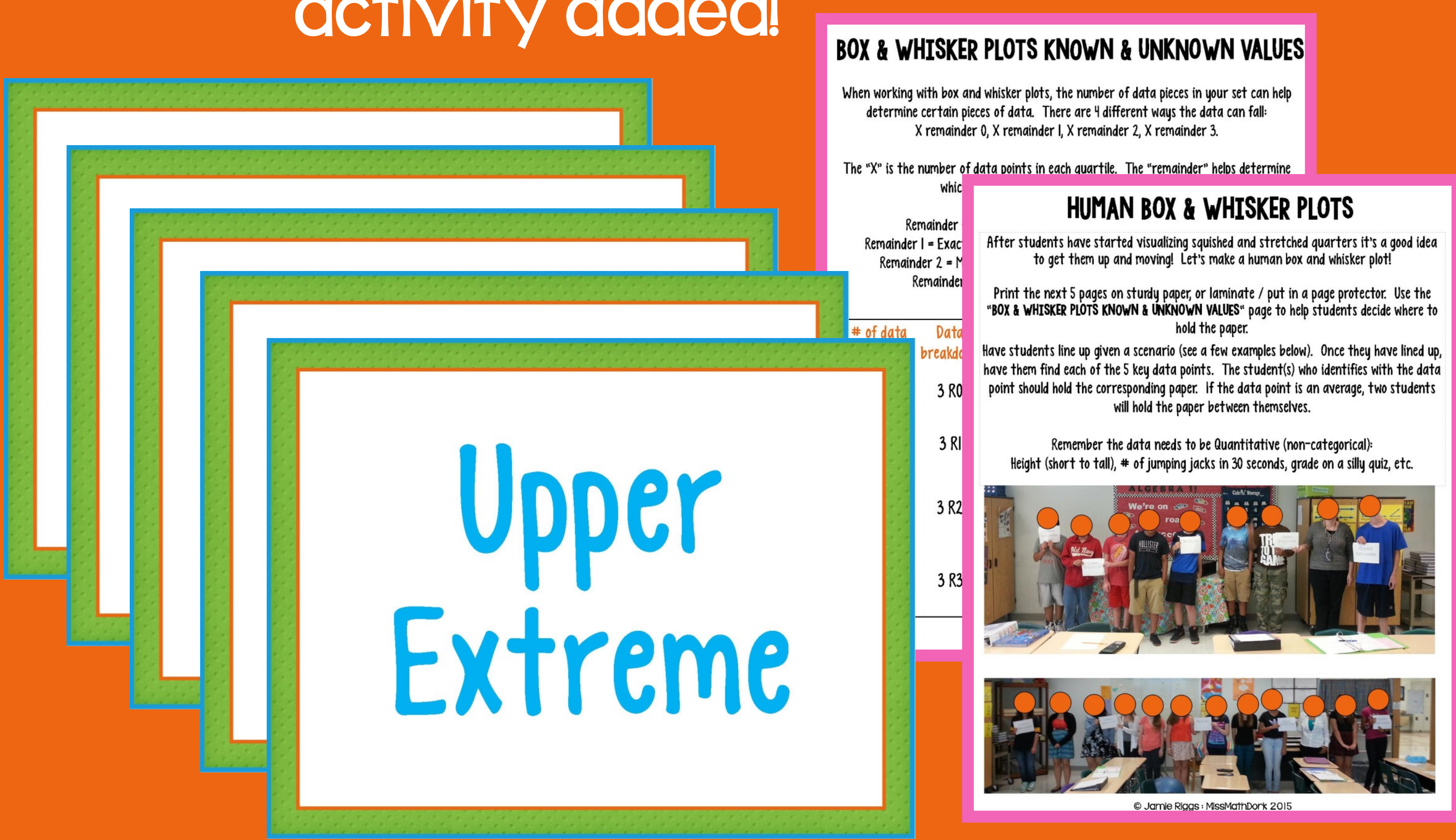
12. In which set of data, is the Median the smallest? Set A the median is 10

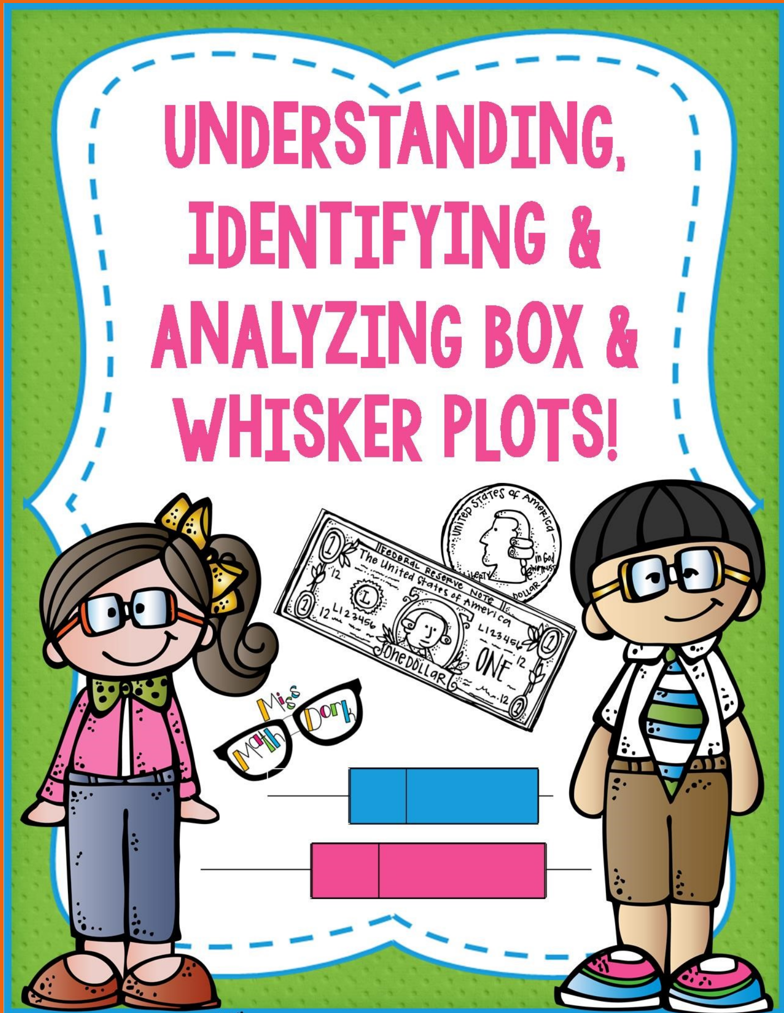
# IDENTIFYING/ANALYZING BOX & WHISKER PLOTS PG 4 KEY

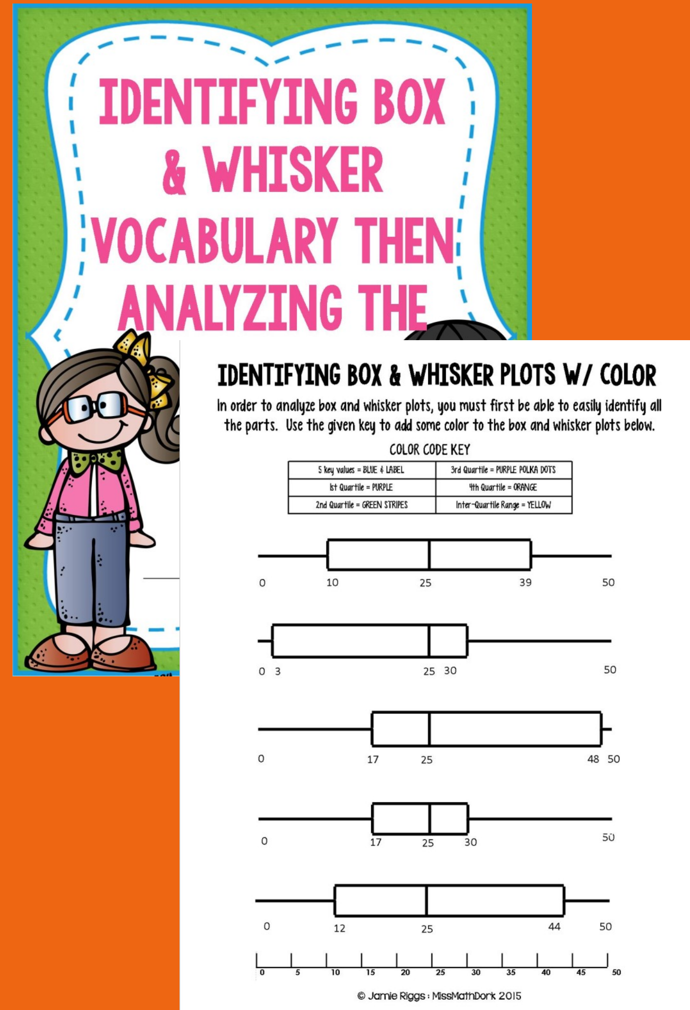
1. How many students make up 25% of Mrs. H's class? 5 students
2. How many students make up 25% of Miss K's class? 6 students
3. How many students make up 25% of Mrs. Ms class? 4 students
4. How many students make up 25% of Mrs. R's class? 5 students
5. Comparing Mrs. H's and Miss. K's classes: Which class has a greater number of student's scoring at least a 90 on their math test? Miss K has 6 while Mrs. H has 5.
6. Comparing Mrs. M's and Mrs. R's classes: Which class has a greater number of student's scoring at most an 80 on their math test? Mrs. M has 4 while Mrs. R has 10.
7. Which class has the highest median score? Miss K's median is 85.
8. Who has the most students scoring at least a 75 on their test? Miss K or Mrs. R? Miss K has 18 students scoring a 75 or more, Mrs. Rigg's only has 15 students scoring a 75 or more.



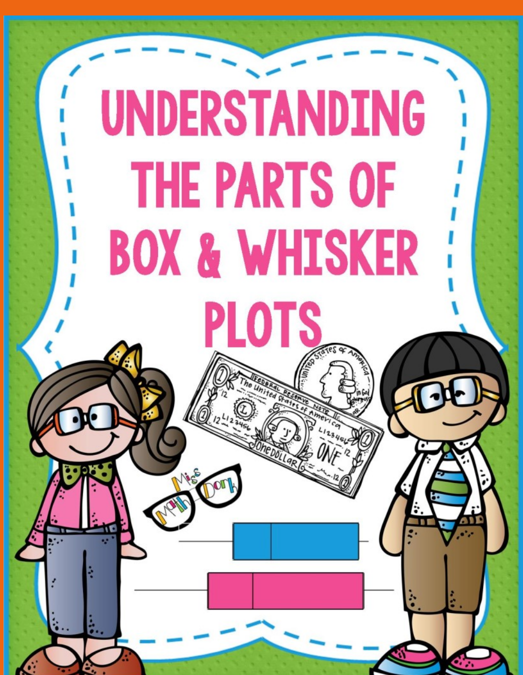

Revamped ! New notes, posters & interactive activity added!

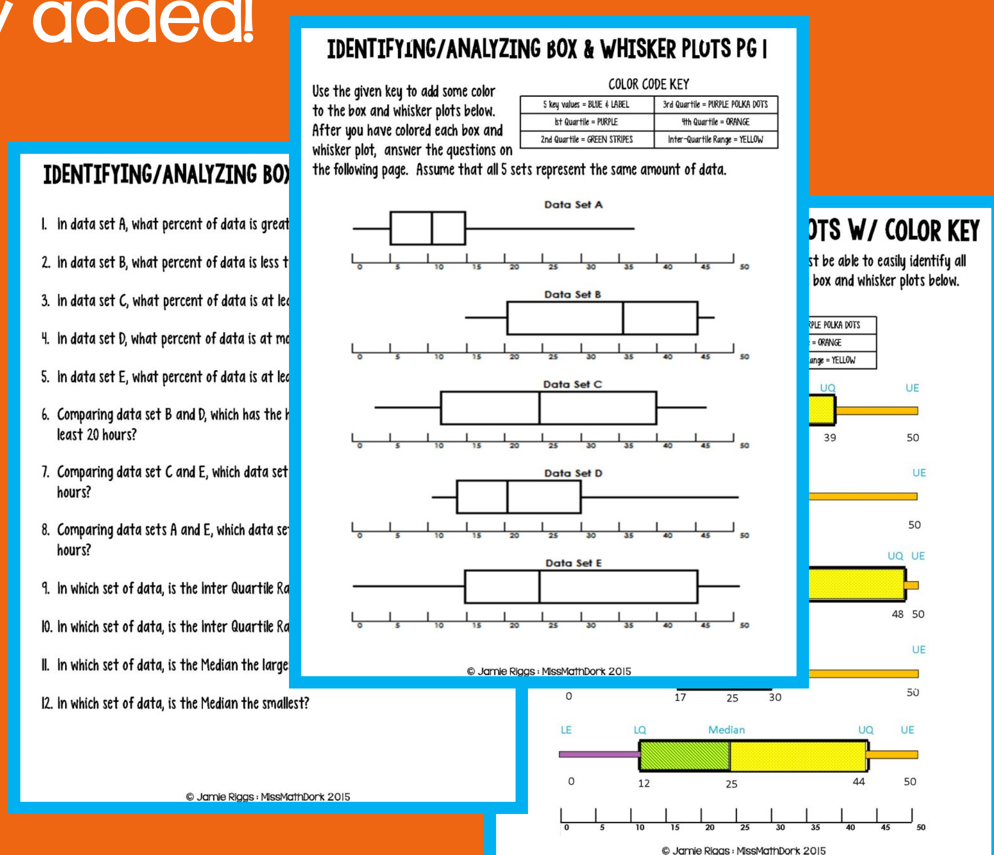





Revamped ! New notes, posters & interactive activity added!



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